3c5570d1-76cf-49d3-b653-cbc1b9827b56

<current>

<request>

You are working on an issue in the 'musicmonk42/VulcanAMI_LLM' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/VulcanAMI_LLM/VulcanAMI_LLM (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2025-11-24T20:34:37.453Z</current_datetime>

Consider the following problem statement:

<problem_statement>
is there anything in the entire plaform not running that should be?

Platform Core

Unified Platform (Worker 67004)
Server process on 0.0.0.0:8080
JWT Authentication
Metrics (Enabled)
Health Checks (Enabled)

Environment & Configuration

UTF-8 stdout/stderr reconfiguration
Environment variables loader (from .env)
OPENAI_API_KEY loader
ANTHROPIC_API_KEY loader
GRAPHIX_API_KEY loader
WorldModelConfig definition

```
File watcher for configs\profile_development.json
Configuration loader from configs\profile_development.json
Runtime override: intrinsic_drives_config.enabled = True
Runtime override: intrinsic_drives_config.approval_required =
False
Runtime override: intrinsic_drives_config.check_interval_seconds
= 120
Runtime override:
intrinsic_drives_config.max_cost_usd_per_session = 2.0
Runtime override: intrinsic_drives_config.max_cost_usd_per_day =
10.0
Runtime override: intrinsic_drives_config.auto_apply_policy
Environment validation


Safety System - Neural Safety

Neural dependencies check (torch v2.9.1+cpu)
Neural dependencies check (scipy v1.13.1)
Neural dependencies check (statsmodels v0.14.5)
SafetyPredictor module
FeatureExtractor module
initialize_neural_safety function
LLM safety validator modules
NeuralSafetyValidator (5 model types on CPU)


Safety System - Domain Validators

DomainValidatorRegistry
CausalSafetyValidator
PredictionSafetyValidator
OptimizationSafetyValidator
DataProcessingSafetyValidator
ModelInferenceSafetyValidator


Safety System - Core Components

ToolSafetyManager
ToolSafetyGovernor
ComplianceMapper (9 standards)
BiasDetector (multi-model ensemble on CPU)
RollbackManager (max_snapshots=100)
AuditLogger (path: safety_audit)
AdversarialValidator (epsilon=0.1)
FormalVerifier
```

```
SecurityNodes (EncryptNode and PolicyNode)
InterpretabilityEngine (device=cpu)


Safety System - Governance

AdaptiveGovernance (lazy-loaded)
EnhancedNSOAligner (lazy-loaded)
SymbolicSafetyChecker (lazy-loaded)
GovernanceManager


Safety System - Constraints

Constraint: energy_conservation (priority: 10)
Constraint: uncertainty_bounds (priority: 8)
Constraint: minimum_confidence (priority: 7)
Constraint: resource_limits (priority: 9)


Safety System - Properties & Invariants

Safety property: basic_safety
Safety property: resource_bounds
Safety property: action_consistency
Invariant: state_consistency
Invariant: system_stability


Hardware Emulation

AnalogPhotonicEmulator (backend: cpu)
HardwareEmulator (noise_std=0.01, noise_type=gaussian)
HardwareDispatcher (2 backends discovered)
Hardware health check thread
HardwareDispatcherIntegration


Semantic Bridge

Semantic Bridge v1.0.0 (Production)
SemanticBridge (production-ready)
ConceptMapper (production-ready, bounded data structures)
EvidenceWeightedResolver (production-ready, bounded data
structures)
PartialTransferEngine (with learning)
TransferEngine (production-ready, bounded data structures)
```

```
DomainRegistry (production-ready, bounded data structures)
RiskAdjuster
CacheManager (1000 MB limit)


Registered Domains

Domain: general (criticality: 0.10 / 0.50)
Domain: safety_critical (criticality: 0.95)
Domain: optimization (criticality: 0.50)
Domain: real_time (criticality: 0.90)
Domain: machine_learning (criticality: 0.70)
Domain: classification (criticality: 0.50)
Domain: generation (criticality: 0.50)
Domain: planning (criticality: 0.50)
Domain: control (criticality: 0.50)
Domain: reasoning (criticality: 0.50)
Domain: perception (criticality: 0.50)


World Model Core Components

CausalDAG (complete implementation)
CorrelationMatrix
CorrelationTracker (method=pearson)
InterventionPrioritizer (refactored)
InterventionExecutor (simulation_mode=True)
PathTracer (refactored with type checking)
EnsemblePredictor (method=weighted_quantile, clustering=auto)
DynamicsModel
InvariantRegistry (refactored with safety and full symbolic
system)
InvariantDetector (refactored with full symbolic system)
SymbolicExpressionSystem (sympy=True)
ConfidenceCalibrator (method=isotonic, n_bins=10)
ModelConfidenceTracker
WorldModelRouter (8 strategies)


World Model Module List

causal_graph
confidence_calibrator
correlation_tracker
dynamics_model
intervention_manager
prediction_engine
```

```
invariant_detector
world_model_core
world_model_router


Meta-Reasoning System

MotivationalIntrospection (6 objectives)
ObjectiveHierarchy (6 objectives)
CounterfactualObjectiveReasoner
GoalConflictDetector
ValidationTracker
TransparencyInterface
objective_hierarchy (lazy loaded)
counterfactual_objectives (lazy loaded)
goal_conflict_detector (lazy loaded)
validation_tracker (lazy loaded)
transparency_interface (lazy loaded)


Self-Improvement System

SelfImprovementDrive (5 objectives)
CSIU enforcement module (safety controls)
CSIU weights persistence
Agent state loader (data\agent_state.json)
Design spec loader (configs\intrinsic_drives.json)
Auto-apply policy (enabled)


Memory System

FAISS (AVX2 support)
Vulcan Persistent Memory v46.0.0
Groth16 SNARK module (falling back to basic implementation)


Reasoning System

Vulcan Reasoning Module (5/5 reasoners available)
causallearn (GES/FCI algorithms)
Symbolic reasoning (from symbolic.reasoner)
Analogical reasoning (spaCy model not loaded - warning)
ToolSelector selection support components
Contextual bandit
```

```
Orchestrator System

VULCAN-AGI Orchestrator (Version 1.0.2)
Agent lifecycle state machine validation
ExperimentGenerator
ProblemExecutor
Self-improvement system
Ray support
ZeroMQ support
No circular dependencies
Bounded memory usage
Thread-safe operations
Proper resource cleanup
Comprehensive error handling


GraphixVulcanLLM System

GraphixVulcanLLM v2.0.2
GraphixTransformer
GraphixVulcanBridge
SafeGeneration
EnhancedSafetyValidator
ExplainableGeneration
HierarchicalContext
CausalContext
CognitiveLoop
LanguageReasoning
GovernedTrainer
SelfImprovingTraining


GraphixExecutor Instances

GraphixExecutor #1 (512d, 6L, 8H, 57 weight tensors)
GraphixExecutor #2 (256d, 6L, 4H, 57 weight tensors)
Model config: 50257 vocab, 512d
Config: max_tokens=128, temp=0.7


Unified Runtime

UnifiedRuntime (Grammar 2.0.0)
Manifest loader (graphix_core_manifest.json)
AIRuntime (providers: openai, mock, default)
ExecutionEngine (max_parallel=10, timeout=300.0s)
GraphValidator (ontology from graphix_core_ontology.json)
```

```
SubgraphLearner (0 patterns)
EvolutionEngine (pop_size=100, cache_size=10000,
diversity_threshold=0.1)
AutonomousOptimizer (Evolution=True, Optimizer=False, NSO=False,
Governance=True)
ExecutionExplainer
RuntimeExtensions
ObservabilityManager (log_dir=observability_logs)
ConsensusEngine (quorum=0.51, approval_threshold=0.66)


Governance Loop Policies

Policy: Resource Safety (ID: safety_001)
Policy: Latency Requirements (ID: perf_001)
Policy: No Harm Principle (ID: ethical_001)
GovernanceLoop


VULCAN Integration

VULCAN World Model integration
VULCAN MotivationalIntrospection re-initialization
VULCAN integration enabled in UnifiedRuntime


Graphix Arena

GraphixArena (3 agents: generator, evolver, visualizer)
GraphixLLMClient (agent: arena-agent, model: gpt-4o,
temperature: 0.01)
DataAugmentor
DriftDetector (dim=128, threshold=0.1, history=5, method=center)
TournamentManager (similarity_threshold=0.8,
diversity_penalty=0.2, winner_percentage=0.1)
NSOAligner (audit database)
SecurityAuditEngine (audit.db)
SecurityAuditEngine (nso_aligner_logs\audit.db)
Slack alerting (#graphix-arena-alerts)
tool_selection.yaml configuration
Protected routes registration


Registry (Flask App)

Rate limiter (Redis: redis://localhost:6379/1)
Redis backend (production-ready)
```

Redis Connection

Redis connection established (for VULCAN main)

Mounted Services

VULCAN mounted at /vulcan
Arena mounted at /arena
Registry mounted at /registry

Endpoints

Safety status endpoint at /safety
/vulcan/docs (OpenAPI documentation)

Total Count: 197 distinct functions/components/modules running
This covers every initialization message, module load, component setup, validator registration, policy addition, and service mount from the startup log.
</problem_statement>

Implement the necessary changes to the repository so that the requirements specified in the <problem_statement> are met.

Ensure that you lint, build and test early and iteratively to validate each code change thoroughly.